1. Item 12 (Most relevant)

> **Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation**

How it is reflected:

- The solution process typically involves more than a single operation and requires students to:
  1. Recall or extract relevant prior information
  2. Perform necessary calculations or comparisons
  3. Integrate results
  4. Reach a final conclusion or judgment

👉 The defining feature of Question 4 is **multi-step reasoning and integration**.

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2. Item 13

> **Use an understanding of the context to guide or expedite the mathematical solving process**

How it is reflected:

- Students must understand the real-world meaning and constraints of wind power generation (e.g. efficiency, feasibility, realistic ranges).
- The final conclusion must be **contextually sensible**, not merely mathematically correct.

👉 Contextual understanding plays a **decisive role** in this question.

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3. Item 10

> **Make sense of, relate and use a variety of representations when interacting with a problem**

How it is reflected:

- Students are often required to relate textual descriptions, numerical data, graphs, or results from previous questions.
- The information sources are interdependent and must be interpreted holistically.

👉 This reflects the ability to **integrate multiple representations**.

4. Item 11

> **Switch between different representations in the process of finding solutions**

How it is reflected:

- Students must move between:
  - Numerical results
  - Graphical or tabular information
  - Real-world interpretations

👉 Representation switching is a **key link** in the reasoning chain.

5. Item 7

> **Manipulate numbers, graphical and statistical data and information**

How it is reflected:

- Students may need to compare, interpret, or further process numerical results;
- The data manipulation involved is not complex but is essential.

👉 Data handling functions as a **supporting skill**.

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6. Item 5

> **Understand and utilise constructs based on definitions, rules and formal systems**

How it is reflected:

- Correct interpretation of mathematical or scientific quantities (e.g. power, rates of change, proportional relationships);
- Reasoning that adheres to basic mathematical rules.

👉 This demonstrates sound understanding of **formal mathematical constructs**.

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7. Item 8

> **Articulate a solution, showing and/or summarising and presenting intermediate mathematical results**

How it is reflected:

- Question 4 often requires students to **explain or justify their reasoning**, not merely state a numerical answer;
- Emphasis is placed on the logical connection between reasoning and conclusion.

👉 Communication and justification are more prominent here.

8. Item 3

> **Implement a given strategy to determine a mathematical solution**

How it is reflected:

- The question often implies a solution pathway (e.g. building on results from earlier questions or comparing alternatives);
- Students are expected to follow this implied strategy.

👉 Strategy implementation is present but not central.

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9. Item 1

> **Perform a simple calculation**

How it is reflected:

- The task may involve limited, straightforward calculations or estimations;
- Computational difficulty is low.

👉 This is an auxiliary rather than focal skill.

① 第 12 项（最相关）

> **Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation**

具体体现：

- 解题过程通常不是单一操作，而是需要：
  1. 回顾或提取已有信息
  2. 进行必要的计算或比较
  3. 综合结果
  4. 得出最终结论或判断

👉 Q4 的核心特征是**多步骤推理与整合**。

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② 第 13 项

> **Use an understanding of the context to guide or expedite the mathematical solving process**

具体体现：

- 学生必须理解风力发电在现实中的意义与限制（如效率、可行性、合理范围）。
- 所得结论需要在现实中"说得通"，而不是形式上正确即可。

👉 情境理解在该题中**起决定性作用**。

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③ 第 10 项

> **Make sense of, relate and use a variety of representations when interacting with a problem**

具体体现：

- 题目往往要求学生将文字说明、数据、图表或前题结论联系起来使用。
- 信息之间存在依赖关系，需要整体理解。

👉 属于**跨信息源的理解与整合能力**。

④ 第 11 项

> **Switch between different representations in the process of finding solutions**

具体体现：

- 在解题过程中，学生需要在：
  - 数值结果
  - 图像或表格信息
  - 现实意义

  之间进行转换。

👉 表征之间的切换是推理链中的关键环节。

⑤ 第 7 项

> **Manipulate numbers, graphical and statistical data and information**

具体体现：

- 学生可能需要对已有结果进行比较、解释或再次处理；
- 数据操作本身不复杂，但不可或缺。

👉 数据处理是支撑性能力。

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⑥ 第 5 项

> **Understand and utilise constructs based on definitions, rules and formal systems**

具体体现：

- 正确理解涉及的数学或科学量（如功率、变化关系、比例）。
- 遵循基本数学规则进行推理。

👉 体现对正式数学概念的掌握。

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⑦ 第 8 项

> **Articulate a solution, showing and/or summarising and presenting intermediate mathematical results**

具体体现：

- Q4 通常需要学生**解释或说明理由**，而不仅仅给出一个数值答案。
- 强调结论与理由之间的逻辑联系。

👉 表达与论证能力的重要性上升。

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⑧ 第 3 项

> **Implement a given strategy to determine a mathematical solution**

具体体现：

- 题目往往隐含了解题路径（如基于前题结果、比较不同方案）。
- 学生需要沿着既定思路完成推理。

👉 策略执行存在，但不是主要挑战。

⑨ 第 1 项

> **Perform a simple calculation**

具体体现：

- 可能涉及少量、简单的计算或估算。
- 计算难度较低。

👉 属于辅助性技能。